DMMST — Version 2 results: our loss variants compared with other models

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1 Setup

We compare our model (a transformer with the survival head of the paper) trained with different losses against eight published survival models. All models are run on the same five datasets under the same protocol:

- 10 random splits of each dataset (60% train, 10% validation, 30% test); results are reported as mean (standard deviation) over the 10 splits;
- settings of every model chosen on the validation split only;
- every model scored by the same code.

Datasets. METABRIC and SUPPORT (real, one event: death); EBMT (real, 4 events); hsa synthetic and DeepHit synthetic (published synthetic data, 2 events). SEER was planned as an additional large real dataset, but because of the updated SEER+ data-access policies we could not obtain access to it.

Our variants.

Name	Losses
S1	L_{PCH} (survival likelihood)
S2	$L_{PCH} + L_{rank}$
S3	$L_{PCH} + L_{mul}$ (multi-event data only)
S4	$L_{PCH} + L_{rank} + L_{mul}$ (multi-event data only)
Full	survival head with all its losses + regression head (L_{MAE} with best guess)

L_{mul} compares different events of the same patient, so S3 and S4 do not exist on single-event data ("—" in the tables).

Regression head. Besides the survival curve, the model can predict one number: the time of the event. For censored patients the true time is unknown, so the "best guess" (from a Kaplan–Meier curve of the training data) is used as the target.

Name	Losses
R1	L_{MAE} with best guess
R2	L_{MAE} with best guess $+L_{rank}$
R3	L_{MM} (mismatch loss; multi-event data only)
R4	$L_{MM} + L_{rank}$ (multi-event data only)
Joint	both heads trained together, every S $\times$ R combination

Predicted times are scored by MAE (mean absolute error, lower is better; in the time unit of each dataset) and Harrell C (how well predicted times rank the patients, higher is better).

Baselines. SurvTRACE (authors' code), DeepHit, RSF, PC-Hazard, DeepSurv, MENSA, Cox, DSM.

Scores. C_{td}: how well patients are ranked by risk (higher is better; 0.5 is random). IBS: integrated Brier score, the accuracy of the predicted survival probabilities (lower is better).

2 Results

Table 1: C_{td}, mean (sd) over 10 splits. Best per dataset in bold.

Model	METABRIC	SUPPORT	EBMT	hsa synth.	DeepHit synth.
Ours S1	0.667 (0.013)	0.615 (0.011)	0.529 (0.013)	0.849 (0.049)	0.743 (0.003)
Ours S2	0.670 (0.016)	0.620 (0.007)	0.535 (0.016)	**0.868** (0.010)	0.743 (0.004)
Ours S3	—	—	0.530 (0.016)	0.859 (0.015)	0.742 (0.003)
Ours S4	—	—	0.537 (0.015)	0.867 (0.009)	0.742 (0.004)
Ours Full	0.672 (0.021)	0.619 (0.007)	0.534 (0.017)	0.852 (0.041)	0.744 (0.004)
SurvTRACE	**0.678** (0.013)	**0.638** (0.008)	0.546 (0.009)	**0.868** (0.008)	0.755 (0.007)
DeepHit	0.677 (0.012)	0.610 (0.008)	0.525 (0.011)	0.865 (0.010)	0.737 (0.006)
RSF	0.672 (0.014)	0.636 (0.007)	**0.549** (0.009)	0.849 (0.009)	0.733 (0.005)
PC-Hazard	0.668 (0.017)	0.627 (0.008)	0.548 (0.010)	0.837 (0.008)	**0.762** (0.003)
DeepSurv	0.640 (0.030)	0.593 (0.007)	0.529 (0.014)	0.842 (0.009)	**0.762** (0.002)
MENSA	0.654 (0.017)	0.596 (0.021)	0.501 (0.019)	0.847 (0.014)	0.570 (0.074)
Cox	0.631 (0.022)	0.567 (0.006)	0.531 (0.014)	0.497 (0.013)	0.591 (0.005)
DSM	0.620 (0.065)	0.602 (0.037)	0.512 (0.013)	0.522 (0.031)	0.653 (0.104)

Table 2: Integrated Brier score (lower is better), mean (sd) over 10 splits. Best per dataset in bold.

Model	METABRIC	SUPPORT	EBMT	hsa synth.	DeepHit synth.
Ours S1	0.179 (0.012)	0.200 (0.004)	0.338 (0.165)	0.075 (0.007)	0.174 (0.005)
Ours S2	0.175 (0.006)	0.199 (0.004)	0.313 (0.153)	0.075 (0.003)	0.171 (0.005)
Ours S3	—	—	0.256 (0.082)	0.074 (0.003)	0.175 (0.005)
Ours S4	—	—	0.233 (0.043)	0.075 (0.003)	0.172 (0.006)
Ours Full	0.177 (0.005)	0.200 (0.003)	0.299 (0.138)	0.076 (0.005)	0.172 (0.007)
SurvTRACE	0.201 (0.033)	**0.193** (0.004)	0.242 (0.018)	**0.071** (0.003)	0.169 (0.006)
DeepHit	0.198 (0.007)	0.217 (0.004)	0.388 (0.159)	0.091 (0.002)	0.254 (0.012)
RSF	0.176 (0.007)	**0.193** (0.003)	0.224 (0.009)	0.087 (0.002)	0.185 (0.005)
PC-Hazard	**0.174** (0.006)	0.194 (0.004)	0.219 (0.010)	0.078 (0.003)	0.171 (0.005)
DeepSurv	0.176 (0.006)	0.195 (0.004)	0.220 (0.008)	0.078 (0.003)	**0.167** (0.005)
MENSA	0.189 (0.012)	0.212 (0.014)	0.331 (0.023)	0.079 (0.003)	0.223 (0.014)
Cox	0.175 (0.006)	0.198 (0.004)	**0.218** (0.009)	0.105 (0.002)	0.218 (0.006)
DSM	0.224 (0.067)	0.205 (0.008)	0.326 (0.024)	0.369 (0.049)	0.232 (0.078)

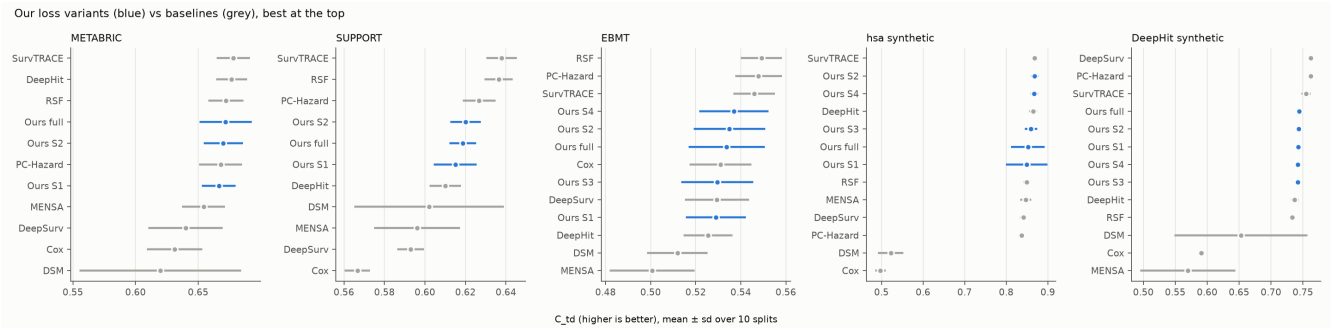


Figure 1: C_{td} of our variants (blue) and the baselines (grey), best at the top of each panel.

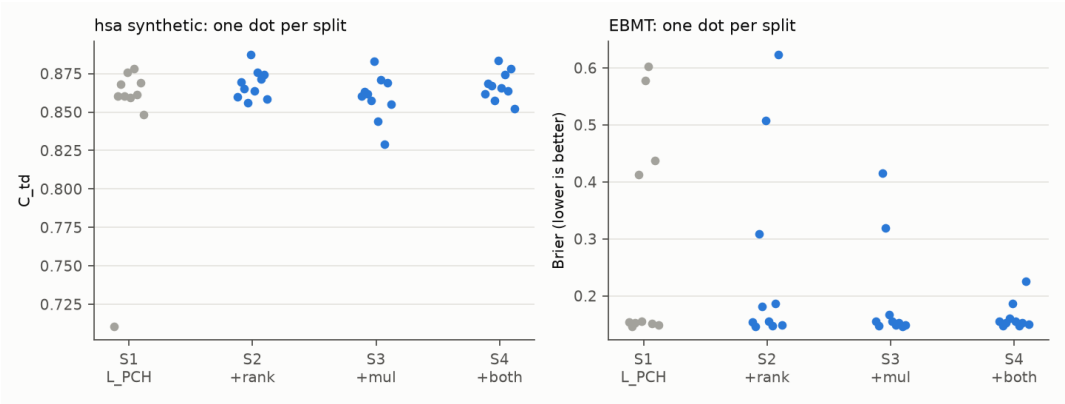


Figure 2: Every split of each loss variant. Left: C_{td} on hsa synthetic. Right: Brier score on EBMT. With L_{PCH} alone (S1) some splits fail; with $L_{rank} + L_{mul}$ (S4) none do.

Regression head

Table 3: Predicted times: Harrell C (higher is better), mean (sd) over 10 splits. Best per dataset in bold. "Joint" is the mean over all S × R combinations.

Recipe	METABRIC	SUPPORT	EBMT	hsa synth.	DeepHit synth.
R1	0.608 (0.018)	0.546 (0.007)	0.510 (0.012)	0.566 (0.021)	0.689 (0.046)
R2	**0.643** (0.012)	**0.592** (0.009)	0.524 (0.014)	0.663 (0.047)	0.704 (0.042)
R3	—	—	0.530 (0.009)	0.663 (0.087)	0.700 (0.035)
R4	—	—	**0.533** (0.011)	0.665 (0.063)	**0.714** (0.003)
Joint	0.626 (0.026)	0.575 (0.023)	0.504 (0.015)	**0.689** (0.125)	0.551 (0.141)

Table 4: Predicted times: MAE (lower is better), mean (sd) over 10 splits.

Recipe	METABRIC	SUPPORT	EBMT	hsa synth.	DeepHit synth.
R1	80.6 (2.8)	613.9 (9.1)	4254 (123)	396.4 (5.8)	38.8 (1.5)
R2	**76.7** (3.2)	**604.6** (9.6)	**4232** (130)	408.7 (96.7)	34.8 (3.2)
R3	—	—	4245 (126)	420.6 (94.9)	35.0 (3.0)
R4	—	—	4247 (120)	**379.8** (62.2)	**33.9** (0.4)
Joint	80.7 (4.7)	612.3 (13.7)	4321 (150)	487.5 (120.6)	44.3 (5.6)

3 Findings

1. **Our model is competitive but not the best.** It is in the upper half on every dataset and ties SurvTRACE on hsa synthetic, but wins no dataset. Our best variant is 0.000–0.018 C_{td} behind the best model.

2. L_{rank} **helps a little on every dataset** (S2 $\geq$ S1). The gain is statistically significant only on hsa synthetic.

3. **The extra losses make training reliable.** With L_{PCH} alone one hsa split collapses and several EBMT splits are badly calibrated. With $L_{rank} + L_{mul}$ (S4) this does not happen: the EBMT IBS drops from 0.338 to 0.233.

4. **Adding the regression head does not hurt survival prediction** when L_{rank} is used ("Full" $\approx$ S2/S4).

5. L_{rank} **also helps the regression head.** R2 ranks predicted times better than R1 on every dataset (e.g. Harrell C 0.566 $\rightarrow$ 0.663 on hsa synthetic) and lowers the MAE on four of five datasets (hsa synthetic is the exception).

6. **Joint training helps predicted times on hsa synthetic but hurts them on DeepHit synthetic** (Harrell C 0.689 vs 0.551), because the checkpoint is chosen by the survival C_{td}, not by the regression score.

7. **Calibration is good** on four of five datasets, and better than SurvTRACE on METABRIC. EBMT is the exception.

8. **EBMT is hard for every model** (C_{td} 0.50–0.55): it has only four categorical features.

4 Conclusion

The ranking losses of the paper do not make the model more accurate than the strongest baselines, but they make it more reliable and better calibrated, and they allow the regression head to be trained together with the survival head at no cost. The results do not support a claim of better C_{td} than SurvTRACE.